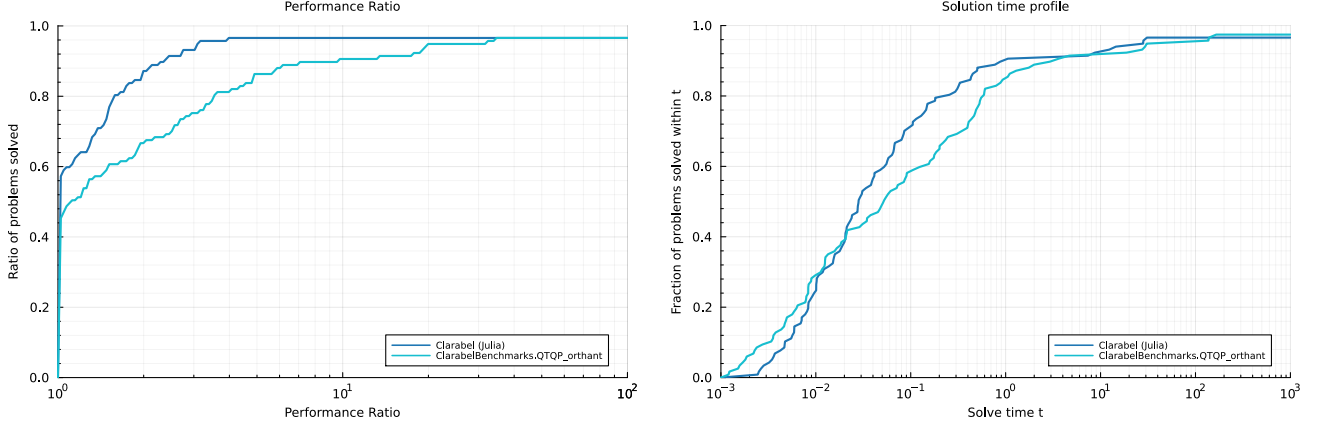


5 Numerical Experiments

5.1 Benchmark problems with quadratic objectives

5.1.1 The Maros-Meszaros test set

Figure 1: Performance profiles for the Maros-Meszaros problem set



(a) Relative performance profile

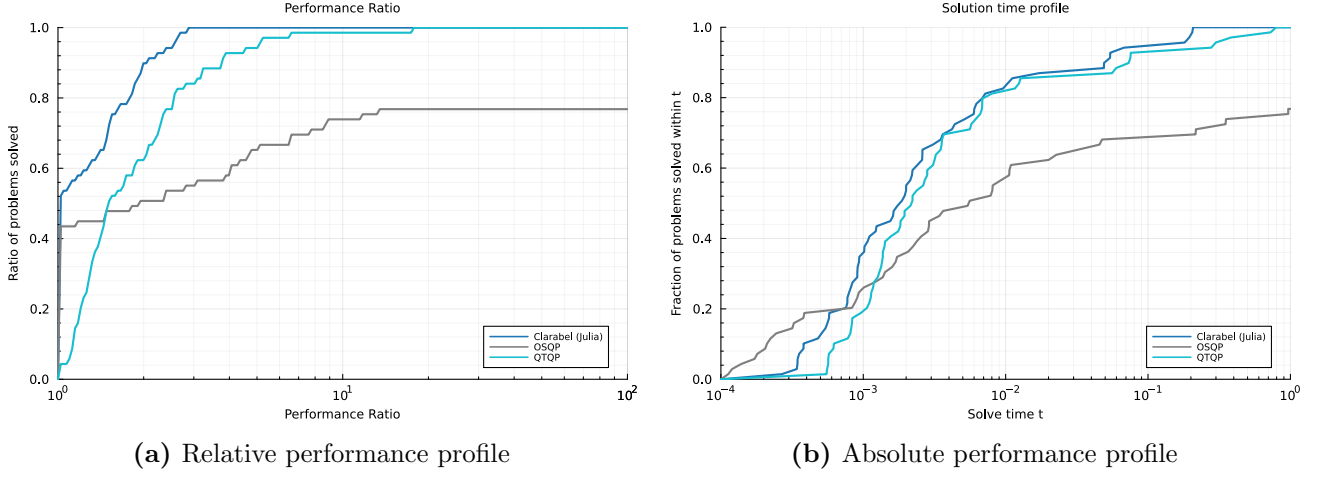
(b) Absolute performance profile

		Clarabel	ClarabelBenchmarks.QTQP_orthant
Shifted GM	Full Acc.	1.0	1.38
	Low Acc.	1.0	2.08
Failure Rate (%)	Full Acc.	3.4	2.6
	Low Acc.	0.9	2.6

(c) Benchmark timings as shifted geometric mean and failure rates

5.1.2 Constrained optimal control

Figure 2: Performance profiles for the optimal control problem set

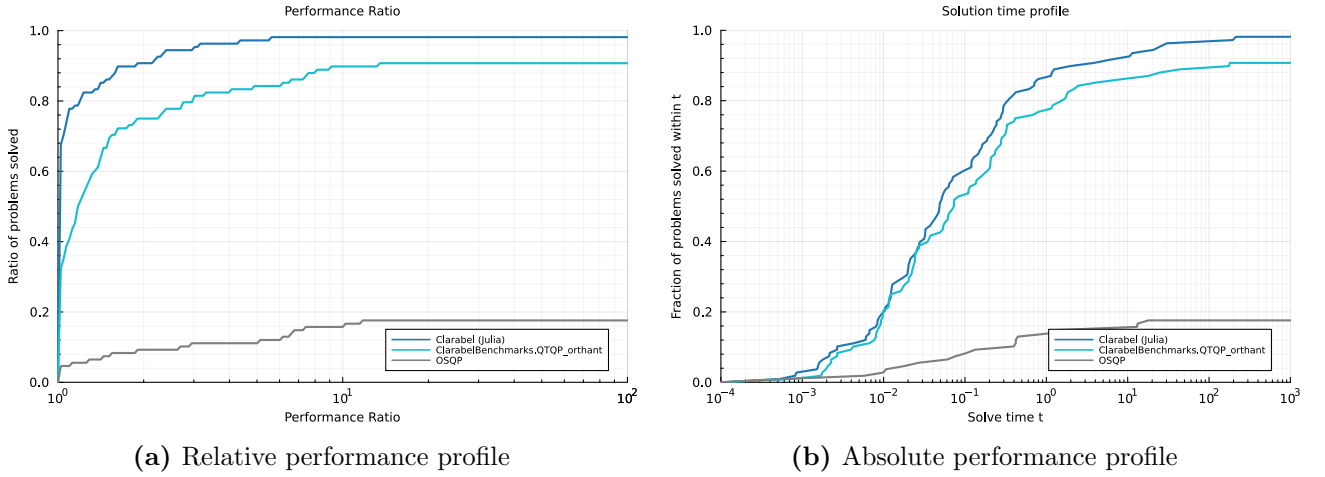


		Clarabel	OSQP	QTQP
Shifted GM	Full Acc.	1.0	173.88	2.18
	Low Acc.	1.0	124.29	2.18
Failure Rate (%)	Full Acc.	0.0	23.2	0.0
	Low Acc.	0.0	18.8	0.0

(c) Benchmark timings as shifted geometric mean and failure rates

5.1.3 NETLIB LP problems

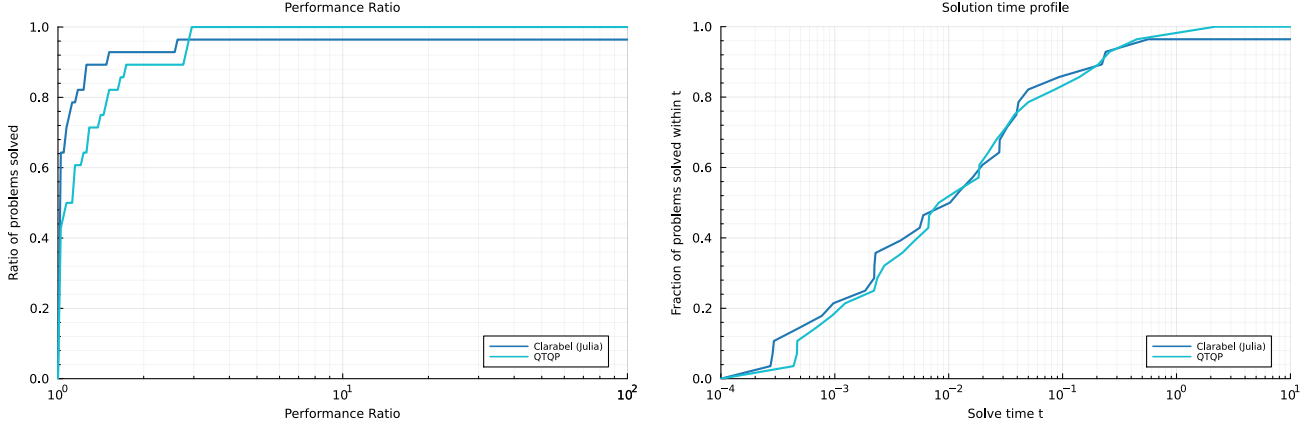
Figure 3: Performance profiles for the NETLIB Feasible LP problem set



		Clarabel	ClarabelBenchmarks.QTQP_orthant	OSQP
Shifted GM	Full Acc.	1.0	2.32	194.8
	Low Acc.	1.0	2.66	133.18
Failure Rate (%)	Full Acc.	1.9	9.3	82.4
	Low Acc.	0.9	9.3	71.3

(c) Benchmark timings as shifted geometric mean and failure rates

Figure 4: Performance profiles for the NETLIB Infeasible LP problem set



(a) Relative performance profile

(b) Absolute performance profile

		Clarabel	QTQP
Shifted GM	Full Acc.	3.12	1.0
	Low Acc.	1.0	1.02
Failure Rate (%)	Full Acc.	3.6	0.0
	Low Acc.	0.0	0.0

(c) Benchmark timings as shifted geometric mean and failure rates

A Detailed benchmark results

Table 1: Solve times and iteration counts for the Maros-Meszaros problem set

Problem	<u>iterations</u>		<u>total time (s)</u>			
	time per iteration(s)		vars.	cons.	nnz(A)	nnz(P)

Table 2: Solve times and iteration counts for the Optimal Control problem set

Problem	<u>iterations</u>		<u>total time (s)</u>		<u>nnz(A)</u>		<u>nnz(P)</u>	
	vars.	cons.	vars.	cons.	vars.	cons.	vars.	cons.

Table 3: Solve times and iteration counts for the NETLIB Feasible LP problem set

Problem	<u>iterations</u>		<u>total time (s)</u>			
	<u>time per iteration(s)</u>					
	vars.	cons.	nnz(A)	nnz(P)		